SIMATS ENGINEERING

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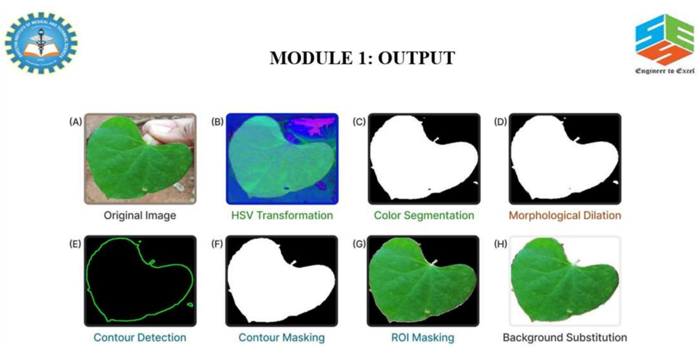
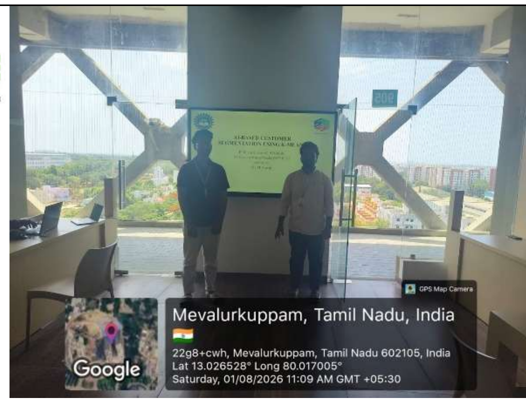
Course Code: **DSA0201** **Slot: C**

Course Name: COMPUTER VISION WITH OPENCV

Course Faculty: Dr. M. MAYURANATHAN

Project Title: **PLANT LEAF DISEASE IDENTIFICATION AND CLASSIFICATION USING OPENCV AND MACHINE LEARNING**
Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)





Project Description

The project "Plant Leaf Disease Identification and Classification Using OpenCV and Machine Learning" is designed to automatically identify and classify diseases in plant leaves from digital images. The system combines OpenCV-based image processing techniques with Machine Learning algorithms to extract important visual features from leaves and predict their disease category.

The input leaf image is first processed using OpenCV to improve its quality and isolate the leaf region from the background. The image undergoes color-space transformation, segmentation, morphological operations, contour detection, contour masking, and Region of Interest (ROI) extraction. These operations help remove unwanted background information and focus only on the leaf area.

After preprocessing, relevant features such as color, texture, shape, and disease-affected regions are extracted from the leaf. These features are then provided to a Machine Learning classification model, which learns patterns from a labeled plant-leaf dataset and identifies whether the leaf is healthy or diseased. Depending on the dataset, the system can classify diseases such as Leaf Spot, Powdery Mildew, Rust, Blight, or other crop-specific diseases.

Student Signature **Guide Signature**